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Mechanical manifestations of rare atomic jumps in dynamic force microscopy

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Abstract
The resonance frequency and the excitation amplitude of a silicon cantilever have been measured as a function of distance to a cleaved KBr(001) surface with a low-temperature scanning force microscope (SFM) in ultrahigh vacuum. We identify two regimes of tip–sample distances. Above a site-dependent critical tip–sample distance reproducible data with low noise and no interaction-induced energy dissipation are measured. In this regime reproducible SFM images can be recorded. At closer tip–sample distances, above two distinct atomic sites, the frequency values jump between two limiting curves on a timescale of tens of milliseconds. Furthermore, additional energy dissipation occurs wherever jumps are observed. We attribute both phenomena to rarely occurring changes in the tip apex configuration which are affected by short-range interactions with the sample. Their respective magnitudes are related to each other. A specific candidate two-level system is also proposed.

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1. Introduction

Atomic resolution with a scanning force microscope (SFM) operated in a dynamic 'non-contact' mode in ultrahigh vacuum (UHV) [1] has been achieved on a variety of surfaces [2]. During imaging, the shift of the resonance frequency, due to interactions with the tip at the cantilever end, is kept constant. The tip oscillation amplitude is also kept constant and large enough to yield a restoring force that prevents the tip from ever sticking to the sample. The tip apex can then sense interactions with surface atoms at its turning point near the sample. The separation between this point and the top surface layer (ignoring tip-induced deformations) defines the nominal tip–sample distance s. Besides imaging, site-specific frequency–distance measurements can be performed using a drift-free, low-noise low-temperature SFM [3]. The force on the tip determined from the frequency shift provides insight into short-range interactions at different sites and thus into mechanisms producing atomic-scale contrast. In the non-contact range this interaction is essentially conservative. However, a site-dependent energy dissipation is observed at closer tip–sample spacings [4–7]. Dissipation caused by interactions is manifested via changes of the excitation amplitude necessary to keep the tip oscillation amplitude constant. The mechanisms leading to distance-dependent atomic-scale contrast in the excitation amplitude are still the subject of controversies [6]. In particular, artifacts may be produced by feedback errors [8, 9]. Nevertheless, a number of experimental observations reveal genuine intrinsic energy dissipation of the order of 0.1 to a few eV per cycle [4–6].

Two apparently distinct dissipation mechanisms of mechanical origin have been proposed [10]. The stochastic

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damping mechanism relates dissipation to the Brownian motion of atoms [11]. Forces exerted between thermally fluctuating sample atoms lead to a net friction force on the moving tip similar to viscous friction on a body moving in a fluid. Model calculations of the energy dissipated per cycle [12] are close to predictions from molecular dynamics simulations extrapolated to experimental frequencies [13], but orders of magnitude below typical values measured close to contact [4, 6].

Values in the experimental range are, however, predicted by the atomic adhesion hysteresis mechanism [14, 15] which postulates that an atom hopping back and forth between tip and sample leads to a force difference between approach and retraction in each oscillation cycle. The area enclosed by the resulting hysteresis loop yields the energy dissipated per cycle. With the presently used cantilever oscillation amplitudes of a few nanometers, the two different forces cannot be extracted because the measured frequency shift only senses the average tip–sample interaction in the two states. Nevertheless jumps in the frequency shift and in the dissipation are predicted to occur at a critical distance below which the position of an atom on the tip or the sample becomes unstable [14]. Actually, the transition between the stochastic damping and hysteresis regimes can be somewhat smeared by thermal fluctuations, as demonstrated in recent simulations [13, 16], as well as by external noise.

The above-mentioned treatments considered atomic hops between the tip and the sample, but hops with a *lateral* component on the tip (or on the sample) can also occur. Except in special symmetry-preserving situations, such hops will also modify the tip–sample interaction, hence change the coupling to the oscillating tip, and thus affect the measured frequency shift and energy dissipation. In the case of an atom hopping between two potential wells, any coupling causing that potential to oscillate back and forth leads to maximum dissipation when the oscillation frequency matches the mean hopping rate. This generic model, originally proposed by Zener [17], has been elaborated further under the name of 'stochastic resonance' and recently invoked in the context of dynamic force measurements [5, 18]. If that rate is nominally low but increases with decreasing tip–sample distance, owing to lowering of the energy barrier between the wells, a maximum in dissipation versus distance curve is expected. Such a trend has, in fact, been observed using a small-amplitude off-resonance mode [5].

The present work is concerned with a somewhat similar situation, but in a different limit that, to our knowledge, has so far not been considered, namely when rare hops occur on the timescale of the measurement and can therefore be directly observed. Actually the hops can happen between configurations involving several atoms rather than a single one. The lowest intervening barrier singles out a *'two-level system'* of the kind postulated to explain anomalous thermal properties of amorphous solids [19]. Direct evidence for individual hopping defects came later from low-temperature observations of switching between two values (telegraph noise) in the electrical resistance of submicrometer metallic wires [20, 21]. We present and discuss analogous mechanical manifestations of such rare hops in low-temperature frequency–distance measurements on an insulating ionic crystal. A small but significant onset of energy dissipation is observed below the distance where jumps appear in the frequency shift. We argue that the underlying hops occur near the tip apex and that the magnitudes of the energy dissipation can be related to the frequency jumps under plausible assumptions. A possible two-level system which might account for our observations is discussed and further measurements are suggested.

2. Measurements

The experiments were performed at 8 K using a home-built ultrahigh-vacuum, low-temperature SFM [22]. Silicon cantilevers with well-defined dimensions were ordered from Nanosensors™. The cantilever used for the present measurements had a bending spring constant $c_b = 40$ N m^{-1} (within a 15% uncertainty mainly arising from the uncertainty in the thickness determination). The cantilever was introduced into the UHV chamber and cleaned by heating to 150 °C for 30 min in order to remove physisorbed water and contaminant layers. This mild treatment is insufficient to remove the thin native silicon oxide covering the cantilever and the tip. A clean flat KBr(001) surface was prepared by cleaving a single crystal in UHV. In order to remove surface charges created in the process, the crystal was heated to 120 °C over 1 h. After cooling to 8 K, the measured resonance frequency of the freely oscillating cantilever was $f_0 = 160\,622.47$ Hz. The resonance frequency shift Δf was measured using a demodulation circuit controlled by a digital phase-locked loop which also adjusts the cantilever excitation [23]. The frequency shift and the excitation voltage V_{exc} necessary to maintain a constant tip oscillation amplitude of $A = 5.8$ nm were recorded simultaneously.

Right on resonance, if $\Delta f \ll f_0$, the interaction-induced energy dissipated per cycle is given by [4]

$$\Delta E = \frac{V_{exc} - V_{exc0}}{V_{exc0}} \frac{\pi c_b A^2}{Q_0}, \tag{1}$$

where the first fraction is the relative change in the excitation voltage and the second fraction is the energy dissipated in the absence of interaction, ΔE_0. The quality factor Q_0 was determined from the width δf of the resonance characteristic (amplitude versus frequency) of the weakly excited lever far from the sample, with

$$Q_0 = \frac{f_0}{\delta f}, \tag{2}$$

$f_0 \pm \delta f/2$ being the frequencies where the amplitude A is reduced to $1/\sqrt{2}$ of its maximum value.

Intrinsic Q_0 values of 10^4–10^5 have been measured at room temperature in UHV for commercially available cantilevers thinner than ours, hence more prone to Q_0 reduction by surface defects [24]. Roughly one order of magnitude higher values have been obtained at low temperatures [25]. We were able to achieve stable operation with an even *higher effective* Q_0 by taking advantage of the Q control provided by our cantilever deflection sensor. In this device, laser light reflected back into a single-mode optical fiber leading out of the cryostat generates an interference pattern in the cavity formed by the fiber end and the backside

of the cantilever [26]. Maximum deflection sensitivity is achieved at positions where the slope of the interference pattern versus fiber–cantilever separation is extremal. Such positions occur on opposite sides of each interference minimum. As recently shown [27], Q_0 is effectively enhanced on one side and reduced on the other side. As in active Q control [28], this effect arises from the photothermal response of the cantilever (partial absorption of light at its end and inhomogeneous diffusion of the generated heat towards its support). More details about these effects will be discussed elsewhere [29]. However, in the context discussed here, it is important to stress that this modification of the Q factor is entirely passive, i.e. is not related to an additional source or sink of energy. The following observations allow us to exclude laser-induced driving under the conditions described here: before switching the sinusoidal excitation voltage on, a possible spontaneous cantilever oscillation remained below the experimental detection limit. If additional energy flew into the cantilever oscillation from the interferometer, one would, on the contrary, expect that the cantilever would oscillate. However, if a finite but sufficiently small sinusoidal voltage was applied to the excitation piezo, a constant steady-state tip oscillation could be maintained. Once the excitation was subsequently switched off, the cantilever oscillation died off over tens of seconds. If the cantilever were still driven by the interferometer, we would expect it to reach a steady state characterized by a nonzero oscillation amplitude.

All measurements reported here were performed with an effective $Q_0 \approx 6.4 \times 10^6$ which corresponds to $\Delta E_0 = 4.1$ meV per cycle. The precision of our digital frequency detector (0.01 Hz) [23] determines the uncertainty in δf. Considering the calibration errors in all quantities entering the dissipated energy by the free cantilever, ΔE_0, as statistically independent, namely $\delta c_b \sim 15\%$, $\delta A \sim 30\%$ (A is, however, held constant to a much better accuracy) and $\delta Q_0^{-1} \sim 40\%$ (according to equation (2)), we obtain a total rms error of 75% in ΔE_0. In view of the approximate nature of the theoretical estimates provided in section 3, the presently achieved compromise between order-of-magnitude accuracy and unprecedented sensitivity to low energy dissipation is sufficient for our purposes.

First, a flat area of the sample was located and zoomed onto. After several contrast changes, stable images were repeatedly obtained at a constant frequency shift of -19.5 Hz [30]. These changes are attributed to KBr molecules being picked up and/or released by the tip apex. Then, measurements of the frequency shift and the cantilever excitation amplitude versus distance were performed above four sites readily identified in the images and labeled in the inset of figure 1(a), namely the minimum (1), the maximum (2) and the bridge sites (3, 4). The latter two sites are crystallographically equivalent on a perfect surface. Differences in the frequency shifts measured above these two sites therefore reflect asymmetries of the tip and/or of the pattern of atomic displacements induced at close distances. In a series of such measurements, the frequency shift and excitation amplitude were measured over 60 ms at each data point while smoothly approaching and retracting the sample to and from the tip during a sweep of typically 30 s. After each pair of measurements at two inequivalent

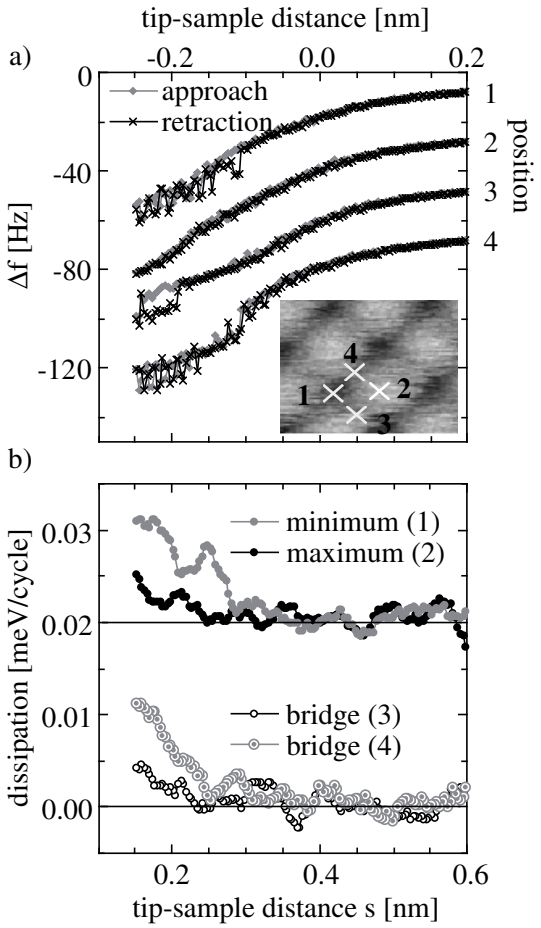


Figure 1. Frequency shift (a) and average dissipated energy per oscillation cycle (b) as functions of the tip–sample distance. The inset shows locations 1–4 where the data were acquired on a small area from a larger undisturbed SFM image recorded at $\Delta f = -19.5$ Hz. The data obtained at different positions are vertically offset by 20 Hz in (a) and by 0.02 meV in (b). Two tip–sample distance scales are shown: the absolute distance s (bottom), obtained from a comparison between experiment and simulations, and the relative distance with respect to the mean sample displacement at which images were recorded (top).

positions, an image was recorded in order to check whether tip changes and/or sample damage had occurred during the preceding measurements. No modifications were observed. The minimum approach distance was then slightly decreased and the sweep was repeated. During all experiments, we checked that the oscillation amplitude remained constant. Additional experimental details are given elsewhere [30, 31]. In particular, attention was focused on larger distances where (apart from noise) unique, reversible frequency–distance data were obtained and no differences were observed between sites 3 and 4 [30]. In the present work we focus on measurements extending to closer distances.

2.1. Frequency versus distance; rare multiple jumps

At all positions, the data (see figure 1) show the typical decrease of the frequency shift towards more negative values with decreasing distance, mainly being due to attractive long-range forces. Short-range forces cause the small differences in

the frequency shift between atomic sites observed around and below the imaging distance. A careful comparison between force–distance data computed from the measurements and from simulations [30] allows us to calibrate the absolute (closest) distances as indicated in figure 1. In order to eliminate the influence of long-range forces, force differences between the two sites were calculated. We estimate the accuracy of the distance scale calibration to be of the order of 0.1 nm. More details about this comparison are given in [30]. In the raw data displayed in figure 1, only noise without obvious jumps is observed at position 2. At sites 1 and 4, similar behaviour is observed above site-dependent critical tip–sample distances of ≈0.3 nm, in particular at the imaging distance. Below the critical distance, multiple jumps occur, the retraction showing different frequency shift values from the approach. At site 3 a single jump occurred near the distance of closest approach (0.15 nm) and the new state was preserved up to 0.22 nm upon retraction, except for a jump to the initial state and back near 0.15 nm. Note that the magnitude of the frequency jump is much larger than the width of the cantilever resonance, 0.025 Hz. The jumps in question reproducibly occurred below the same site-dependent critical distances within about 0.1 nm until a permanent tip change occurred. One should, however, not expect to exactly reproduce the same characteristics when using a different tip. Nevertheless qualitatively similar phenomena are often observed in both static [32] and dynamic measurement modes, during frequency versus distance measurements as well as during imaging where they appear in the form of sudden contrast changes attributed to 'tip changes' (see, for instance, [6, 31]).

This peculiar site dependence indicates that the observed jumps are not artifacts. Judging from tabulated empirical radii of the interacting species (K^+, Br^-, Si, O), repulsive forces are expected to become noticeable at the small distances where jumps are observed. It is therefore remarkable that images acquired after the approach and retraction measurements remained essentially identical, i.e. were unaffected by any of the observed jumps. This indicates that no permanent tip or sample modification occurred but suggests that the jumps reflect sudden changes of the tip–sample interaction at distances where short-range forces become significant and sensitive to asymmetries of the tip apex. More precisely, the invariant appearance of the images implies that the force on the tip may be modified for a while, but recovers its initial distance dependence before the next image can be measured (recall that the frequency shift is the same during approach and retraction at distances larger than 0.3 nm for position 1 in figure 1). At least for the system considered here, there exists a distance range which is intermediate between that of truly non-contact operation and that of truly irreversible plastic deformation.

We first discuss possible effects of jumps in the interaction on the measured frequency shift and only later characterize the underlying 'tip changes'. Whenever jumps are observed, the frequency values appear to lie between two limiting envelopes. The top one looks like the smooth continuation of the dependence measured above the critical distance below which jumps are observed. This suggests that a new state characterized by a somewhat stronger attractive interaction corresponding to the lower envelope becomes available. In order to justify this interpretation, consider the expression for the frequency shift given by the following integral of the force $F(z)$ on the tip over the oscillation period $\tau = 1/f$ [33] where z is the momentary tip position:

$$\Delta f(s) = -\frac{f_0}{c_b A}\int_0^\tau F(\bar{z} + A\cos\omega t)\cos\omega t \frac{\mathrm{d}t}{\tau} \qquad (3)$$

where $\omega = 2\pi/\tau$, $\bar{z}$ is the equilibrium position of the cantilever and $s = \bar{z} - A$ is the distance on the bottom scale of the figures. This expression is valid in the limit $c_b A \gg F$, i.e. $\Delta f \ll f_0$, which is appropriate for our measurements, and strictly speaking, if $F(z)$ is single-valued and time-independent. $F(s)$ can then be reconstructed from the measured frequency shift starting from a large distance where both Δf and $F(z)$ can be assumed to vanish. This has been done earlier above the distances at which jumps appear [30]. Vibrations of atoms or ions are typically much faster than the oscillation period of the cantilever and therefore average out in $F(z)$, and hence in Δf. However, a sudden tip change can lead to a modified $F(z)$ which persists for a while until the next change. If only a few tip changes occur during the measurement time t_m at each data point, equation (3) can be generalized by replacing τ by t_m, and weighting the force $F_i(z)$ corresponding to state i by the time spent in that state. This generalization is justified if those time intervals are much longer than the oscillation period. This is in fact the case: although a few frequency values lie between the two envelopes, the larger fraction of successive data points jumps from one to the other envelope. This is only possible if the mean time between tip changes is comparable to the measurement time (60 ms) which encompasses almost 10^4 oscillation periods.

2.2. Energy dissipation versus distance

Independently of the origin of the tip changes, it is clear that a force difference between approach and retraction during an oscillation cycle causes energy dissipation, as in the adhesion hysteresis mechanism [14, 15]. For the rare jumps in tip–sample interaction inferred here, this will be true for at least some of the cycles. However, since the mean time between jumps is much longer than the tip oscillation period, we expect a small dissipation compared to previous experiments [4, 6]. There, dissipation measured at close distances could roughly be explained by one atomic jump per oscillation cycle [14], causing a force difference of ∼0.1 to 1 nN in a hysteresis loop extending over ∼0.1 nm.

Figure 1(b) shows the additional energy dissipated per cycle in close proximity of the sample. Zero corresponds to the energy dissipated by the cantilever far from the sample, i.e. $\Delta E_0 = 4.1$ meV. The conversion was done using equation (1). The experimental noise has been reduced by applying a running average over 10 points to the data measured during approach simultaneously with the frequency shift. Residual slow variations, which cause seemingly unphysical negative values of ΔE could not be completely eliminated. Nevertheless distinctly stronger rises in dissipation towards lower distance are revealed at positions 1 and 4. These rises occur in the same distance range where the jumps in the frequency data appear, suggesting that the two phenomena are related. As expected from the long mean time between tip changes, the energy dissipation is small. In this case,

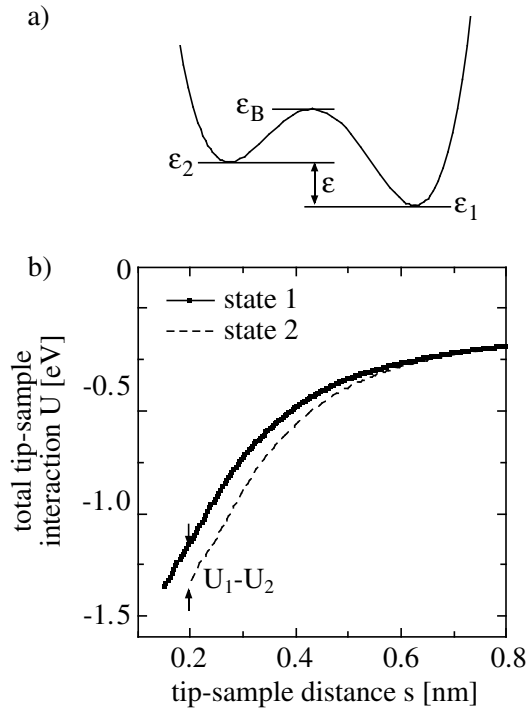


Figure 2. (a) Schematic double well potential showing the relevant energies. (b) Total tip–sample interaction energy versus absolute distance s at position 1 for state 1 obtained by integrating inwards the force $F_1(s)$ extracted from the frequency versus distance points and their upper envelope (squares). The corresponding energy for state 2 (dashed black curve) was approximated by taking $F_2(s) = F_1(s - \Delta s)$ (see text).

mechanisms leading to weak energy dissipation in every cycle, e.g. stochastic damping, might also become significant. The weak increases observed at positions 2 and 3, where no jumps are observed, support this possibility. Hereafter we focus on the stronger energy dissipation that appears related to the jumps.

3. Discussion

3.1. Two-level system near the tip apex

By analogy with the telegraph noise observed in the resistance of metallic wires, we attribute the observed jumps to an atom or a group of atoms hopping near the tip apex in a double well potential between two local minima (or energy levels) ε_1 and ε_2, separated by an energy barrier of height ε_B, as sketched in figure 2(a). In the following we will focus on the behavior at position 1. Assuming thermal activation, the hopping rates are given by

$$\gamma_{1\to2} = \gamma_0 e^{-(\varepsilon_B-\varepsilon_1)/k_BT},$$
$$\gamma_{2\to1} = \gamma_0 e^{-(\varepsilon_B-\varepsilon_2)/k_BT}, \qquad (4)$$

where γ_0 is the attempt frequency. Note that the hopping rate is smaller from the state of lower energy. At low temperature, tunneling cannot be excluded as an alternative or parallel hopping mechanism, but for the following discussion it is only important that both rates are slow and that their ratio obeys the detailed balance condition

$$\gamma_{1\to2}/\gamma_{2\to1} = e^{-\varepsilon/k_BT} = \tau_2/\tau_1, \qquad \varepsilon = \varepsilon_2 - \varepsilon_1, \qquad (5)$$

which means that thermal equilibrium of the two-level system within a state 1 or 2 is established between successive hops. The asymmetry ε may in general depend on the tip–sample distance. However, the roughly constant mean jump rate observed in a finite range, e.g. between 0.15 and 0.3 nm at position 1 (see figure 1(a)), can at first sight be explained by hops in directions predominantly parallel to the surface, for which the parameters $\varepsilon_B - \varepsilon_2$ and ε are expected to be only weakly dependent on distance. Moreover, the ratios of the numbers of points on the upper and the lower frequency envelopes is of order 1 implying that $\tau_1 \approx \tau_2$, i.e. $|\varepsilon| \sim k_BT = 0.7$ meV in the above-mentioned range. Furthermore, the fact that most successive data points jump between the upper and lower Δf versus s envelopes implies that $\gamma_{1\to2} \approx \gamma_{2\to1} \approx 1/t_m = 17$ Hz.

The energy barrier for the hopping processes can be estimated in the distance range where jumps are observed. Inserting $\gamma_{1\to2} \approx \gamma_{2\to1} \approx 17$ Hz into equation (4) gives $\varepsilon_B - \varepsilon_2 = 17$ meV assuming thermal activation with $\gamma_0 = 10^{12}$ Hz and $T = 8$ K. Since the energy barrier is a logarithmic function of the rate, uncertainties in the two rates and the attempt frequency lead to only small errors on the energy barrier. Furthermore, the relevant local temperature of the tip may be higher than the temperature of the sample since the cantilever is heated near its end by the laser used for the interferometric deflection detection system [22]. The noise spectrum of a similar cantilever was measured with a spectrum analyzer at a large tip–sample distance. Fitting yielded a noise temperature of 40 ± 10 K. Taking $T = 40$ K we estimate an energy barrier of $\varepsilon_B - \varepsilon_2 = 85$ meV. On the other hand, the tip apex is much closer to the cold sample than to the cantilever and may be cooled through radiative heat transfer, so that the local temperature and energy barrier relevant for the hops could be lower.

The discussion following equation (3) implies that, after a given hop, the final state must be preserved over many oscillation cycles. Therefore a finite barrier must exist at all distances. In other words, the postulated two-level system is always present near the tip apex, but can be modified by interactions with the sample at short distances.

3.2. Distance dependence of potential asymmetry ε

We now argue that the distance dependence of the potential asymmetry ε is closely related to the difference of the tip–sample interaction energies in the two states. On the one hand, the energies ε_1, ε_2 are determined by interactions of the hopping ions with surrounding atoms or ions of the tip, but also with nearby ions of the sample at short distances. On the other hand, the tip–sample interaction energy in each of the two states,

$$U_i(s) = \int_s^\infty F_i(s')\,ds', \qquad i = 1, 2 \qquad (6)$$

contains the second above-mentioned contribution, but also other ones from ions which do not directly participate in the

hops. Consequently the distance dependence of $\varepsilon = \varepsilon_2 - \varepsilon_1$ and of $U_1 - U_2$ arises from the same subset of ions. Both differences mainly come from short-range interactions, in particular from those ions which hop by a distance of the order of a lattice spacing. To put the preceding argument on firmer ground, note that for our purposes each interaction energy can be represented as a sum over pairs of ions belonging to the tip *and* to the sample, respectively (direct term)[5]. An indirect (relaxation) contribution comes from ions belonging to the tip *or* to the sample which interact with the former ones. If the displacements of those relaxed ions are proportional to the forces exerted by the ions interacting directly, the *change in* ε, between s and a distance $s_l \approx 1$ nm beyond which short-range forces become negligible, should be equal to the corresponding change in the total tip–sample interaction energy difference $U_1 - U_2$. Note that ε_1, ε_2 and ε_B are measured from a common reference energy which corresponds, for example, to separated atoms or ions. Thus, ε can be finite whereas $U_1 \to U_2$ at large distances, so that

$$\begin{aligned}\varepsilon(s_l) - \varepsilon(s) &= [U_1(s) - U_1(s_l)] - [U_2(s) - U_2(s_l)] \\ &= U_1(s) - U_2(s),\end{aligned} \tag{7}$$

as long as displacements can be superposed linearly. If some nearby ions also undergo significant displacements in the course of a hop or at close distances, this relation becomes approximate, but is still expected to give a reasonable estimate of the change in ε.

From the discussion of the hopping rates in the preceding section (section 3.1) we concluded that $\varepsilon_1 \approx \varepsilon_2$, i.e. $\varepsilon \approx 0$ at distances s where frequency jumps are observed. The preceding argument therefore indicates that in the same range

$$\varepsilon(s_l) \approx U_1(s) - U_2(s). \tag{8}$$

In order to estimate $U_1 - U_2$, note that, in a particular state, equation (3) implies that, at a given distance in the jump range, Δf is determined by the corresponding force law in the interval $[\bar{z} - A, \bar{z} + A]$. Therefore, we expect $F_2 \neq F_1$ beyond the onset of the observed jumps, although $F_2(s)$ cannot be reconstructed just because jumps no longer occur, i.e. $\tau_1 \gg \tau_2$. As a reasonable guess we take $F_2(s) = F_1(s - \Delta s)$ with $\Delta s = 0.05$ nm chosen so as to best fit the lower frequency envelope. Integrating $F_1 - F_2$ inwards from the distance s_l where the two coincide, we obtain the interaction energies plotted in figure 2(b) which differ by 0.20 eV in the range where the frequency jumps are observed. ε quickly becomes much larger than $k_B T$ beyond the onset of the observed jumps, so that state 1 is then predominantly occupied. At our measuring temperature, the crossover from $\tau_1 \sim \tau_2$ to $\tau_1 \gg \tau_2$ occurs over a narrow distance range around $s = 0.3$ nm.

3.3. Relation between interaction energy difference and dissipation

Next we argue that the same energy difference $U_1 - U_2$ roughly accounts for the rise and maximum value of the energy dissipation at position 1 shown in figure 1(b). The mechanical energy W_m dissipated at a given measurement point is given by the irreversible work done by the oscillating tip (velocity v) which can be expressed as

$$W_m = -\int_0^{t_m} F v \, dt, \tag{9}$$

where the integral extends over the measurement time t_m and F is the projection of the force on the tip along its trajectory (irrespective of whether transverse forces are acting). An oscillation cycle in which a $1 \to 2$ hop occurs at z_j makes a contribution

$$\int_{z_j}^{z_m} (F_1 - F_2)\, dz > 0, \tag{10}$$

where $z_j \geqslant s = \bar{z} - A$, and $z_m = \bar{z} + A$ can be replaced by the smaller distance z_l beyond which $F_1 = F_2$. Indeed A is larger than z_l, whereas the probability of a backward jump during the same cycle is negligible. If for simplicity we assume that the lower barrier $\varepsilon_B - \varepsilon_2$ has no significant distance dependence, the following cycles are, on average, unlikely to contribute to equation (9). Any $2 \to 1$ hop contributes a negative amount (energy gain), but the net energy dissipation is nevertheless positive. Indeed $1 \to 2$ hops are less likely to occur at large distances. The corresponding z_j values are therefore distributed over shorter distances than those for $2 \to 1$ hops, so that the positive contributions prevail. Setting $z_j = s$ for the $1 \to 2$ hops and ignoring $2 \to 1$ hops, one obtains an upper bound on the net dissipation equal to the difference between the two curves in figure 2, i.e. 0.20 eV around $s = 0.2$ nm at position 1. The corresponding bound on ΔE, namely 0.020 meV, obtained by dividing that value by the number of cycles sampled per jump, $f_0 \cdot t_m = 9450$, is only twice the energy dissipation at 0.2 nm in figure 1(b) at position 1. In view of the uncertainties in the calibration of Q_0 and A in equation (1), this agreement indicates that the order of magnitude of the observed dissipated energy per cycle is consistent with the proposed mechanism.

3.4. Candidate two-level systems

It will be useful in the future to compare the preceding estimates of the energy barrier and asymmetry ε with values computed for specific candidate two-level systems. Our measurements provide some guidance in that respect. First, the estimate of $U_1 - U_2$ at position 1 turns out to be of the same order as the difference between U_1 and the interaction energy extracted from the unique frequency–distance dependence at position 2. Such small differences of a few tenths of electronvolt cannot arise from configurations in which the tip and the sample bear net charges [34]. Next, because sample material is likely to be picked up by the tip, a small neutral KBr cluster is a plausible model for the tip apex. Computations for a model system consisting of a cubic KBr cluster with a ⟨111⟩ axis perpendicular to a KBr(001) slab were recently reported [30]. In the range where no jumps occur, the magnitude and the distance dependence of site-dependent differences between frequency shifts could be approximately reproduced by computations assuming a K^+-terminated tip.

Movies (available at stacks.iop.org/Nano/18/395503) illustrating calculated displacements of nearby ions upon approach and retraction of such a tip can be viewed

[5] Coulomb and, especially, polarization terms are more difficult to treat but, in the end, also produce a rapidly decaying total contribution, provided that both tip and sample remain neutral [34]. We therefore ignore this difficulty in the present context.

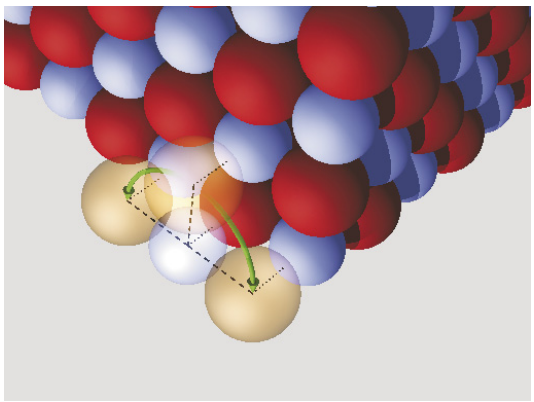


Figure 3. Schematic cluster model of tip apex expected to exhibit the two-level system behavior inferred from the measurements: a 'KBr molecule' hopping by pivoting around one of its partners bound to a counterion on one of the stable {100} facets of a KBr cubic cluster. Large (red and gold) spheres: Br^- ions, small (blue) spheres: K^+ ions.; the two partners of the molecule are highlighted by different hues. Arrows illustrate two possible $1 \rightarrow 2$ hops of the Br^- ion of the molecule (see text).

as supplementary material[6]. At distances closer than those previously investigated [30] these movies (available at stacks.iop.org/Nano/18/395503) reveals that the tip apex is deflected aside even above high-symmetry sites. Nevertheless the model tip recovered its initial configuration upon retraction. Thus there was no evidence for inequivalent tip states separated by a small energy barrier at large distances for that model.

An alternative model which is expected to exhibit the required inequivalent states consists of a 'KBr molecule' (composed of one alkali and one halogen ion) on a stable {100} facet near the apex of a cubic KBr cluster where predominantly the Br^- ion hops (figure 3). We expect that the adsorption and barrier configurations of the KBr molecule will be qualitatively similar to those of a NaCl molecule on the MgO[001] surface computed by Shluger and coworkers [35]. The alkali (halogen) of the molecule preferentially binds to a substrate anion (cation). The lowest barrier corresponds to a rotation of the molecule parallel to the surface around one of those ions. On the (001) surface the initial and final configurations are equivalent, but are no longer so near the tip apex. In particular, if the rotated ion binds to a site of lower coordination, the process corresponds to a $1 \rightarrow 2$ hop with $\varepsilon_1 < \varepsilon_2$ (weaker binding, i.e. less negative energy in state 2). Two such hops are indicated by the arrows in figure 3 in the extreme case of a KBr molecule adsorbed very close to the tip apex. The Br^- ion can hop to an adjacent K^+ site, located either at the apex or at the edge. The second hop is purely lateral with respect to the cluster axis, but will nevertheless give rise to a slightly modified force–distance dependence, except for particular relative orientations and positions of the sample.

[6] See the multimedia files online illustrating deformations which accompany the approach and retraction of a K^+-terminated ⟨111⟩-directed KBr model tip against a K^+ and a Br^- ion in the top layer of a KBr(001) slab. Ion displacements are exaggerated by a factor of 3 for easier visualization. The movies (available at stacks.iop.org/Nano/18/395503) were generated from snapshots taken at increments of 0.02 nm between 6.1 and 2.6 nm using the relaxation procedure described in [30].

Relaxation effects, which are especially important near low-coordinated sites [35], will certainly affect quantitative values of binding energies and energy barrier(s). To our knowledge, the only experimentally available information about KBr on KBr(001), derived from the observation of layer-by-layer homoepitaxy down to 120 K, is that the relevant diffusion barrier is below 0.1 eV [36]. This upper bound is close to the estimate (85 meV) obtained in section 3.1 assuming thermal activation with an effective temperature of 40 K.

Finally we can tentatively explain why ε_2 tends towards ε_1 at small distances and why this first occurs at position 1. At small enough distances the molecule also interacts with the sample. In a situation like that depicted by the arrow pointing towards the tip apex in figure 3, the onset of repulsion could stop the drop of ε_2 initially caused by attraction of the Br^- ion bound aside the tip apex in state 2. In the configuration corresponding to state 1 only attraction would be acting on that ion. Evidence for a site-dependent onset of short-range repulsive forces comes from the changes of curvature apparent in figure 1(a). Indeed, for large oscillation amplitudes, the frequency shift is approximately proportional to $\sqrt{UF}$ at the turning point close to the sample [37]. The inflection point in Δf must therefore occur between the inflection points in the force F, which signals the onset of repulsion, and the inflection point of the interaction energy U (minimum of F). Figure 1(a) shows that, at position 2, the inflection point in Δf occurs at a smaller distance than the inflection point in the upper envelope at position 1. Presumably no jumps are observed at position 2 because ε is still much larger than $k_B T$ down to 0.15 nm.

We also relate this work to recent investigations concerned with tip-induced atom jumps and their observable manifestations in dynamic force microscopy. One set of publications deals with atomistic simulations of hops involving specific surface atoms towards the tip and (eventually) back to the surface either on semiconductor [38–40] or on ionic crystal samples [41–43] and with their influence on distance-dependent images and dissipation. So far, only one of those publications [40] presented a systematic comparison between density functional based simulations for a large number of plausible tip models and short-range forces extracted from dynamic force measurements at 80 K. Trevethan and coworkers focused on the influence of temperature, resonance frequency and detection electronics by combining the master equation approach proposed earlier [16] with a system of equations describing the coupled dynamics of the cantilever and of the various detection and control circuits ('virtual atomic force microscope') [9, 46, 47]. In other recent publications [44, 45] an improved kinetic Monte Carlo algorithm was used to solve the same equations and to investigate in detail the lateral manipulation of surface defects on a MgO(001) surface. In particular, the controlled hop of a specific O nearest neighbor of an O vacancy was predicted with a high probability if the tip was oscillated over a small area between the vacancy and the O neighbor in question. If the oscillating cantilever was released from a safe imaging distance, and the distance controller allowed to reach a frequency shift setpoint for which a jump was predicted at zero temperature, the recorded frequency shift and excitation amplitude showed concomitant fingerprints (a

downwards jump and a peak, respectively) at slightly larger distances at and below 100 K.

The phenomena described in the present article are different in several respects: (i) the approach and retraction measurements were performed at closely spaced points with the distance controller switched off; (ii) multiple frequency jumps which occur below a certain site-dependent distance can only be observed if the underlying infrequent hops are separated by time intervals comparable to the measurement time at every point; (iii) the postulated hops occur near the tip apex rather than on the sample surface. Moreover, essentially all simulations have so far assumed that a hop and a backward one occur in every oscillation cycle, or that a single hop occurs once (permanent tip change due to a picked surface atom). In view of our findings, it would be worthwhile to extend the master equation based approach to intermediate situations where atomic-scale configuration changes occur in only some of the oscillation cycles and to confirm our simple estimates in the limit where such changes occur on the measurement time scale.

4. Conclusions and outlook

In conclusion, we have shown that jumps in the frequency shift correlated with a rise in energy dissipation below a site-dependent distance can be traced back to rare hopping events close to the apex of the SFM tip. Being modified by short-range interactions with the sample, individual hops cause jumps in the total interaction energy which are in turn manifested in the above-mentioned phenomena. Their common origin allows us to estimate their respective magnitudes. We propose that a KBr molecule executing specific hops near the corner of a small KBr cluster picked from the sample might explain the low energy barrier and distance-dependent asymmetry of the double well potential underlying those phenomena. A validation of the proposed scenario by computer simulations is of course desirable.

On the experimental side, more precise information concerning the statistics of the jumps could be obtained by monitoring the output of the frequency detector over a longer time at a series of distances. The analysis of the stress and temperature dependence of the telegraph noise recorded in the electrical resistance of small disordered conductors [20, 21] provides valuable guidance in that respect. The applied stress plays the same role as the force exerted on the tip in our experiments. A study of the temperature dependence would in principle allow us to distinguish thermal activation from phonon-assisted or direct tunneling. This might soon become possible with the few currently available UHV SFM set-ups which have the required extreme distance stability.

In the context of force microscopy, telegraph noise attributed to thermally activated jumps between metastable states has first been detected on ionic crystals in aqueous solution using a static near-contact mode [32]. Assuming a Boltzmann probability distribution, the statistics of the jumps in cantilever deflection recorded at different distances allow a reconstruction of the potential energy of the tip, provided that it is single-valued and that the relevant energy barriers are not much greater than $k_B T$. Because the potential energy includes tip–sample interactions, as well as the harmonic potential of the cantilever, this technique is useful if both contributions are comparable. This requires a cantilever with a spring constant matched to the interactions to be investigated. In contrast, atomic-scale dynamic force measurements like ours rely on relatively stiff cantilevers, but are restricted to UHV and low-drift conditions and require a more complex analysis in view of the large oscillation amplitude (compared to the minimum tip–sample separation).

Note finally that one can imagine situations intermediate between ours and those considered earlier [13, 14, 16, 18]: changes in tip (or sample) configurations which do not necessarily occur in every cycle, but nevertheless are influenced by (and not only reflected in) the tip–sample interaction, hence neither random nor infrequent. The resulting frequency jumps might in fact be induced in a more or less controlled fashion, e.g. by means of a judicious external perturbation rather than by random noise. To our knowledge, no adequate theoretical framework exists to analyze dynamic force measurements under such conditions. Hopefully our work will stimulate further investigations along those lines.

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